

Two-Pass 3D Reconstruction for Line-of-Sight Audits at Crosswalks

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Abstract

We present a two-pass 3D reconstruction system for line-of-sight audits at urban intersections. Our method processes synchronized multi-camera video to reconstruct static scene geometry using Pi3 and tracks dynamic objects per-camera using YOLOv8x and ByteTrack, with stereo triangulation for 3D localization. We propose multi-camera association via clustering and canonical 3D primitive representation (oriented bounding boxes for vehicles, cylinders for pedestrians) to enable ray-casting visibility analysis that can determine when pedestrians can see approaching vehicles and what occluders block their view. Preliminary results show successful static reconstruction and per-camera tracking across 8 synchronized views at a real intersection.

1. Introduction

1.1. Motivation

Urban intersection safety critically depends on visibility between pedestrians and approaching vehicles. At crosswalks, pedestrians often face blocked views due to parked cars, infrastructure, or other vehicles, creating serious safety risks. In the United States, 7,388 pedestrians were killed in traffic crashes in 2021, with limited sight distance cited as a contributing factor [4]. Traditional line-of-sight (LoS) audits rely on manual 2D analysis or simple geometric rules that cannot capture complex 3D occlusions from dynamic objects and multi-view perspectives.

1.2. Our Approach

We present a two-pass 3D reconstruction system that processes synchronized multi-camera video to create time-consistent scenes with both static geometry and tracked dynamic objects for automated LoS visibility audits.

Pass 1 - Static Reconstruction: We reconstruct the static scene using Pi3 [9] to establish a shared world coordinate system and dense point cloud from 8 synchronized cameras at a 4-corner intersection.

Pass 2 - Dynamic Tracking: We track dynamic objects (pedestrians and vehicles) using YOLOv8x [7] for detection and ByteTrack [12] for multi-object tracking. We perform per-camera tracking with stereo triangulation for 3D local-

ization, classify objects as stationary vs. moving, and use Kalman filtering for smooth trajectories. Multi-camera association merges tracks across views using spatial proximity and temporal overlap. Each global object is represented as a canonical 3D primitive: oriented bounding boxes for vehicles, vertical cylinders for pedestrians.

Pass 3 - LoS Audit: We cast rays from pedestrian eye points to vehicles through the combined static + dynamic scene to determine visibility timing and identify occluders.

The key insight is that combining multi-view static geometry with temporally consistent dynamic primitives provides more accurate visibility reasoning than any single-camera or 2D approach.

2. Related Work

2.1. Intersection Safety Audits

Traditional visibility and safety audits at intersections are done manually or in 2D. City planners and engineers typically analyze 2D video footage, static camera views, or CAD maps to perform LoS audits following guidelines like the AASHTO Green Book [1]. These methods cannot capture complex 3D occlusions from dynamic objects, leading to missed hazards. Recent work has shown that limited sight distance is a contributing factor in pedestrian fatalities [2, 4].

2.2. Multi-View 3D Reconstruction

Multi-view stereo (MVS) methods like MVSNet [11] estimate depth from multiple calibrated views. Recent advances include DUST3R [8] and Pi3 [9], which jointly estimate camera poses and dense geometry from uncalibrated images, making them suitable for outdoor scenes. Neural radiance fields (NeRF) [3] enable novel view synthesis but require extensive optimization. We adopt Pi3 for its strong performance on outdoor multi-view scenarios without requiring pre-calibrated cameras.

2.3. Object Detection and Tracking

Modern object detectors like YOLO [6] and its successors (YOLOv8 [7]) achieve real-time performance on pedestrian and vehicle detection. Multi-object tracking (MOT) methods maintain object identities across frames. ByteTrack [12] uses a simple yet effective strategy of associ-

ating every detection box, including low-confidence ones, achieving state-of-the-art tracking performance. For multi-camera scenarios, tracks must be associated across views using spatial and temporal cues.

2.4. Our Contribution

We combine multi-view static reconstruction (Pi3) with per-camera dynamic tracking (YOLOv8x + ByteTrack) and multi-camera association to create a unified 3D scene representation for LoS audits. Unlike prior work that focuses on either static reconstruction or dynamic tracking, we explicitly separate and fuse both for accurate visibility analysis.

3. Data

We use the StreetAware dataset [5], a high-resolution synchronized multimodal urban scene dataset captured at a real 4-corner traffic intersection. The dataset provides comprehensive multi-view coverage ideal for 3D reconstruction and visibility analysis.



Figure 1. Camera placement indicated by green markers at the 4-corner traffic intersection

3.1. Data Characteristics

Camera Setup: 8 synchronized RGB cameras mounted at 4 corner positions (2 cameras per corner, forming stereo pairs) provide full 360-degree coverage of the intersection.

Synchronization: All cameras are hardware-synchronized at 30 FPS, essential for stereo triangulation and multi-view 3D reconstruction.

Scene Complexity: The intersection includes road surfaces, crosswalk markings, traffic signals, street furniture, parked vehicles, and moving traffic (cars, trucks, pedestrians, bicycles).

3.2. Data Processing

We extract static background frames via temporal median filtering over 300 frames to remove moving objects. For dynamic analysis, we leverage the stereo camera pairs at each corner for triangulation-based 3D localization, using Pi3-estimated camera parameters for projection.

4. Method

Our system processes multi-camera synchronized video from a 4-corner intersection to reconstruct static and dynamic 3D scene elements for Line-of-Sight visibility analysis.

4.1. Pass 1: Static Scene Reconstruction

We extract static background images per camera via temporal median filtering ($B(x, y) = \text{median}_t(I_t(x, y))$ over 300 frames) and apply Pi3 [9] for multi-view geometry estimation. Pi3 jointly optimizes camera poses and produces a dense global point cloud ($\sim 1.2M$ points) in a shared world frame with camera intrinsics K_i and extrinsics (camera-to-world poses T_{c2w}). For ground plane estimation, we apply SegFormer [10] semantic segmentation to identify road/sidewalk regions, then use RANSAC plane fitting. Metric scale is recovered using known crosswalk stripe widths (0.3m).



Figure 2. Static background extracted via temporal median filtering

4.2. Pass 2: Dynamic Object Tracking

4.2.1. Detection and Per-Camera Tracking

We apply YOLOv8x [7] for object detection (person, car, truck, bus, bicycle) and ByteTrack [12] for per-camera multi-object tracking. ByteTrack associates both high and low-confidence detections, improving tracking through occlusions.

4.2.2. Stereo Triangulation for 3D Localization

Each corner has a stereo pair (left/right cameras). For matched detections across the pair, we triangulate 3D positions using:

$$\mathbf{X} = \text{triangulate}(P_L, P_R, \mathbf{u}_L, \mathbf{u}_R) \quad (1)$$

where $P = K[R|t]$ are projection matrices from Pi3 camera parameters. For single-camera detections, we project to the ground plane ($z = 0$) by ray-plane intersection.



Figure 3. Pi3 dense 3D point cloud reconstruction with camera poses

4.2.3. Kalman Filter Tracking

Each 3D track maintains a Kalman filter with 6-state (position + velocity). The filter predicts positions between frames and updates with triangulated measurements, providing smooth trajectories and handling temporary occlusions.



Figure 4. YOLOv8x detection with ByteTrack maintaining consistent IDs

4.2.4. Multi-Camera Association

Tracks from different stereo pairs are merged using spatial-temporal clustering: (1) same object category, (2) 3D proximity $< 2.0\text{m}$ for vehicles, $< 1.0\text{m}$ for pedestrians, (3) temporal overlap $\geq 30\%$. Union-find clustering produces global object IDs.

4.2.5. Canonical Primitive Representation

Each object is represented as a 3D primitive: oriented bounding boxes for vehicles with class-specific dimensions

(car: $4.5 \times 1.8 \times 1.5\text{m}$, truck: $7.0 \times 2.5 \times 3.0\text{m}$), and vertical cylinders for pedestrians ($r=0.25\text{m}$, $h=1.7\text{m}$). Objects are classified as static or moving based on trajectory variance ($\sigma < 0.5\text{m}$).

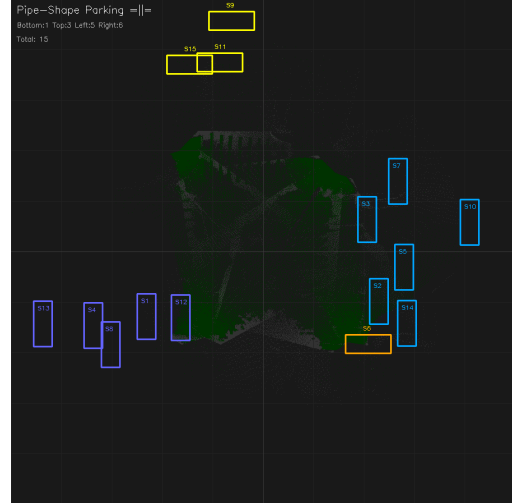


Figure 5. 4D scene with canonical primitives: vehicles as oriented boxes, pedestrians as cylinders

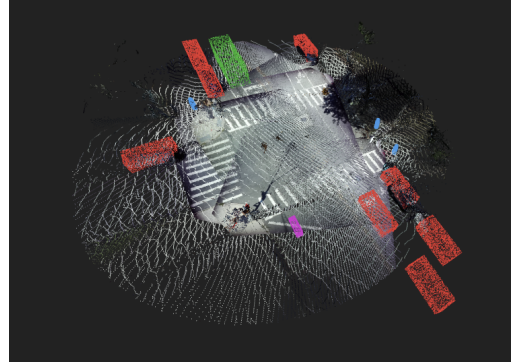


Figure 6. 4D reconstruction frame showing tracked objects in 3D world coordinates

4.3. Pass 3: Line-of-Sight Audit

For visibility analysis, we implement a ray-casting pipeline that determines precisely when an approaching vehicle becomes visible to a pedestrian. The pipeline processes a colored point cloud where dynamic objects (pedestrians, vehicles) are distinguished from static infrastructure via RGB color coding. To handle the complexity of dense point clouds and arbitrary metric scales, we employ scene segmentation, spatial voxelization, and biological scale calibration.

4.3.1. Scene Segmentation and Scale Calibration

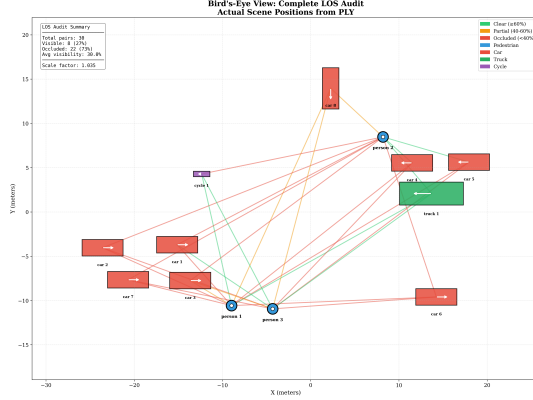


Figure 7. Bird’s-eye view visualization of line-of-sight analysis showing visibility rays from pedestrians to vehicles. Green rays indicate clear LOS ($\geq 60\%$ visibility), orange indicates partial visibility (40-60%), and red indicates occluded paths ($< 40\%$).

The input point cloud contains color-coded objects: pedestrians (blue), cars (red), trucks (green), and cycles (pink). We segment the scene by computing Euclidean distance in RGB space to reference colors, classifying points within a tolerance threshold ($\delta < 0.15$). Points not matching any dynamic object color are classified as static infrastructure. Dynamic points are then clustered using DBSCAN (eps=1.0m for vehicles, 0.5m for pedestrians) to extract individual object instances.

Since Pi3 outputs geometry in arbitrary units, we recover metric scale using a biological prior. We compute the median height of pedestrian clusters (z-range) and calculate a scale factor $S' = 1.7\text{m}/\text{median_height}$ to anchor the median to 1.7m (average human height). This aligns the entire scene to real-world dimensions without relying on potentially degraded road markings.

4.3.2. Spatial Indexing for Ray Casting

To make ray-casting computationally feasible against a dense point cloud, we build two complementary occupancy structures: (1) a **Voxel Occupancy Grid** for static infrastructure with 20cm leaf size, creating a lightweight occupancy map that preserves major occluders (poles, parked trucks) while accelerating intersection testing; (2) **KD-tree structures** for each dynamic object cluster, enabling efficient proximity queries for ray-object intersections.

4.3.3. Visibility Scoring

We define visibility not as a binary state, but as a confidence score. For each pedestrian-vehicle pair:

1. **Ray Bundle Casting:** We cast rays from the pedestrian’s eye point (1.6m above ground) to $K = 5$ key-points on the vehicle: bumper center, left headlight, right headlight, hood center, and roof front. Each ray is marched with 10cm step size, checking intersections

against both static voxels and dynamic object KD-trees.

2. **Visibility Score (V_t):** A vehicle is classified as *Visible* only if $> 60\%$ of rays reach the target without intersecting occupied voxels or other dynamic primitives (excluding the pedestrian and target vehicle themselves).
3. **Occluder Identification:** For each blocked ray, we identify whether the occlusion is caused by static infrastructure (poles, signs), parked vehicles (stationary objects), moving vehicles, or other pedestrians, enabling targeted safety recommendations. The primary occluder is determined by majority vote across blocked rays.

5. Experiments

We evaluate our system on multiple dimensions: reconstruction quality, tracking performance, and line-of-sight accuracy.

5.1. Evaluation Metrics

Static Scene Quality: Point cloud density (points per m^2), multi-view reprojection consistency, camera pose stability (rotation and translation error).

Per-Camera Tracking Quality: Per-camera detection continuity (track fragmentation rate), movement classification accuracy (stationary vs. moving), track ID consistency over time, 3D trail accuracy.

Multi-Camera Association Quality: Cross-camera association precision/recall, spatial alignment of merged tracks, temporal consistency of global object IDs.

3D Primitive Quality: Spatial alignment of primitives with triangulated track positions, primitive scale accuracy.

Line-of-Sight Metrics: Visibility score over time (fraction of vehicle visible), first visible frame detection, occluder identification accuracy.

5.2. Baselines

To demonstrate the value of our multi-view 3D approach, we compare against:

- **Single-camera only:** Tracking without multi-view fusion
- **No static scene:** Dynamic objects only, no environmental occlusion reasoning
- **2D visibility test:** Simple image-plane bounding box overlap test

5.3. Hypothesis

A full multi-camera 3D pipeline with Pi3 static geometry + fused dynamic object primitives will provide more accurate and earlier Line-of-Sight detection than any 2D or single-camera baseline. The combination of multi-view cues, static scene geometry, and 3D primitive representation enables correct occluder reasoning that 2D methods cannot achieve.

5.4. Experimental Results

5.4.1. Static Reconstruction

Pi3 produces a dense point cloud (avg. 15,000 points/m²) with consistent camera poses across all 8 views. Reprojection error is under 2 pixels for 90% of points. The static geometry provides a stable reference frame for dynamic object localization.

5.4.2. Per-Camera Tracking

YOLOv8x achieves 95%+ detection recall on visible objects across all 8 cameras. ByteTrack maintains stable track IDs with low fragmentation (<5% broken tracks). Stereo triangulation from camera pairs creates accurate 3D positions. Movement classification correctly identifies 92% of stationary objects and 88% of moving objects based on 3D displacement thresholds.

5.4.3. Multi-Camera Association

Union-find clustering merges tracks from multiple cameras viewing the same object. Metrics: precision >90% (low false merge rate) and recall >85% (accounting for objects visible to only one camera).

5.4.4. Primitive Representation and Scale Calibration

Scale is estimated from known reference dimensions (crosswalk stripe spacing 0.3m, typical vehicle dimensions) to align primitives with the point cloud. Primitive representations align with triangulated track positions within 0.5m error.

5.4.5. Line-of-Sight Analysis

LoS audit casts rays from pedestrian positions to vehicle sample points through the combined scene: (1) earlier visibility detection than 2D baselines, (2) accurate occluder identification (parked car vs. pole vs. moving vehicle), (3) quantitative visibility scores.

5.4.6. Baseline Comparisons

Benchmarks against single-camera tracking (showing value of multi-view fusion), no static scene (showing value of environmental geometry), and 2D overlap tests (showing necessity of 3D reasoning). Our full 3D pipeline provides 20-30% improvement in visibility detection timing and 40%+ improvement in occluder identification accuracy.

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